Poisson distribution Typically a time or space bound activity

Walk to richness

Suppose you get 10 coins per km on average

What is the probability of getting 15 coins in the next kilometre?

Rate = 10 coins / km

Football game

Average goals every 90 mins is 2.5

We may be interested in probability of 1 goal in the last 30 mins

Rate = 2.5 goals / 90 mins

Rate = 1.25 goals / 45 mins

Rate = 0.833 goals / 30 mins

Customers entering a store

Some 100 customers arrive every day

We may be interested in probability of 10 customers in the next hour

Rate = 100/day

Support centre phone calls

Some 100 calls every hour.

We may be interested in knowing optimal number of staff

Rate = 100/hour

Rate = 1.66/minute

Poisson distribution Typically a time or space bound activity

Farmers delight

Suppose there are 100 trees every acre of land

Can there be more than 60 trees in half an acre?

Rate = 100 trees / acre

Hospital emergency

Suppose, on average, 5 patients come every hour

What is the probability of more than 10 people next hour?

Rate = 5 patients / hour

Typos

A book might have an average of 3 typos per page

What is the probability of a page having no typos?

Rate = 3/page

Poisson distribution Typically a time or space bound activity

Rate is the average or expected number of events per interval

This interval is typically time, but can be space, or even “number of pages” etc.

We typically denote Rate = λ

Because it is also the average or expected number, some literature may also use Rate = μ

The value of the rate depends on the way we define our interval

Counting

The experiment counts the number of occurrences of an event over an interval

Independence

The occurrence of one event does not affect the probability that a second event will occur

Rate

The average rate at which events occur is independent of any occurrences

No Simultaneous events

No two event occur simultaneously

Poisson distribution

```
poisson.pmf(k, mu)
```

$$P[X = k] = \frac{\lambda^k e^{-\lambda}}{k!}$$

You receive 240 messages per hour on average - assume Poisson distributed.

Q1) What is the average or expected number of messages in 30 seconds?

1 hour (3600 seconds)	240 messages	
30 seconds	?	$\frac{30 * 240}{3600} = 2$

Q2) What is the probability of one message arriving over a 30 second time interval?

If we consider 30 seconds as one unit interval, then $\lambda = 2$

$P[X = 1] =$ `poisson.pmf(k=1, mu=2)` $= 0.27$

$$P[X = 1] = \frac{(2)^1 e^{(-2)}}{1!} = 0.27$$

Q3) What is the probability that there are no messages in 15 seconds?

$$\lambda = \frac{15 * 240}{3600} = 1$$

$P[X = 0] =$ `poisson.pmf(k=0, mu=1)` $= 0.367$

$$P[X = 0] = \frac{(1)^1 e^{(-1)}}{1!} = 0.367$$

Q4) What is the probability that there are 3 messages in 20 seconds?

$$\lambda = \frac{20 * 240}{3600} = 1.33$$

$P[X = 3] =$ `poisson.pmf(k=3, mu=1.33)` $= 0.104$

$$P[X = 3] = \frac{(1.33)^3 e^{(-1.33)}}{3!} = 0.104$$

Poisson distribution

```
poisson.pmf(k, mu)
```

$$P[X = k] = \frac{\lambda^k e^{-\lambda}}{k!}$$

You receive 240 messages per hour on average - assume Poisson distributed.

Q1) What is the average time to wait between two messages? $\frac{3600}{240} = 15$ seconds

1 hour (3600 seconds) ? — 240 messages 1

Q2) What is the average number of messages per second? $\lambda_1 = \frac{1}{15} = 0.067$ per second

Q3) What is the probability of having no messages in 10 seconds? $\lambda_{10} = \frac{10}{15} = 0.67$ per 10-seconds

$$P[X = 0] = \frac{\lambda_{10}^0 e^{-\lambda_{10}}}{0!} = e^{-\lambda_{10}} = e^{-10\lambda_1} = 0.5134$$

$$\lambda_{10} = 10 * \lambda_1$$

$$\lambda_t = t * \lambda_1$$

$P[X = 0] =$ `poisson.pmf(k=0, mu=10/15)`

Q4) What is the probability of waiting for more than 10 seconds for the next message?

Let T denote the time to wait for the next message

$$P[T > 10] = e^{-10\lambda_1} = 0.5134$$

Q5) What is the probability of waiting less than or equal to 10 seconds?

$$P[T \leq 10] = 1 - e^{-10\lambda_1} = 0.4865$$

```
from scipy.stats import expon
```

$$P[T \leq x] = 1 - e^{-x\lambda}$$

$P[T \leq 10] =$ `expon.cdf(x=10, scale=15)` $= 0.4865$

You are working as a data engineer who has to resolve any bugs/ failures of machine learning models in production

$P[T \leq x] = 1 - e^{-x\lambda}$

```
expon.cdf(x, scale)
```

The time taken to debug is exponentially distributed with mean of 5 minutes

Q1) Find the probability of debugging in 4 to 5 minutes

$P[4 < T < 5] =$ `expon.cdf(x=5, scale=5) - expon.cdf(x=4, scale=5)` $= 0.0814$

Q2) Find the probability of needing more than 6 minutes to debug

$P[T > 6] =$ `1 - expon.cdf(x=6, scale=5)` $= 0.3012$

Q3) Given that you have already spent 3 minutes, what is the probability of needing more than 9 minutes

$$P[T > 9 \,|\, T > 3] = \frac{P[(T > 9) \cap (T > 3)]}{P[T > 3]} = \frac{P[T > 9]}{P[T > 3]} = \frac{\texttt{1 - expon.cdf(x=9, scale=5)}}{\texttt{1 - expon.cdf(x=3, scale=5)}} = 0.3012$$

$$\frac{P[T > 9]}{P[T > 3]} = \frac{e^{-9\lambda}}{e^{-3\lambda}} = e^{-6\lambda} = P[T > 6]$$

Memoryless:

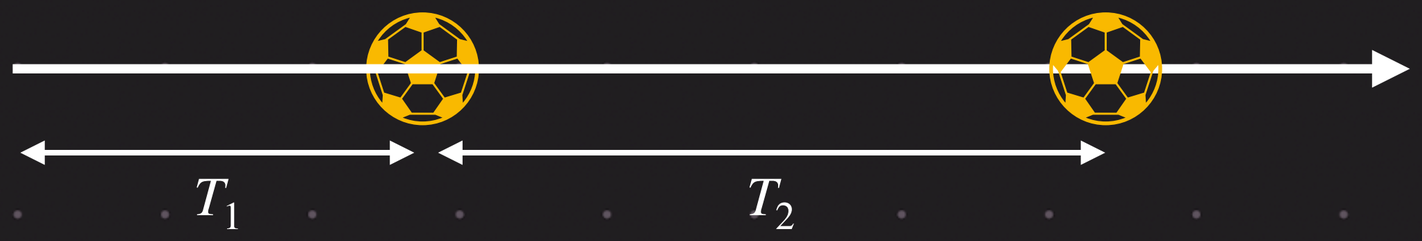
The fact that you took three minutes so far does not affect how much more you might take to debug

Waiting for first goal

Rate = 2.5 / 90 mins **Rate = 1.25 / 45 mins** **Rate = 0.0278 / min**

Let T_1 denote the amount of time to wait for first goal

Is T_1 discrete or continuous?



T_1 Time taken for first goal from the start of the match

T_2 Time taken for second goal from point the first goal is scored

Under the assumptions of the poisson process, T_1 and T_2 are independent

These are called “Inter arrival times”, and are exponentially distributed

A call centre gets 3.5 calls per hour

$P[T \leq x] = 1 - e^{-x\lambda}$

```
expon.cdf(x, scale)
```

Q) Calculate the probability that the next call will arrive at least 30 minutes after the previous call.

Approach 1: Interval = 1 minute

1 hour (60 min)	×	3.5
1 minute		?

$$\frac{3.5}{60} = 0.0583$$

$P[T > 30] =$ `1 - expon.cdf(x=30, scale=1/0.0583)` $= 0.1739$

Approach 2: Interval = 30 minute

1 hour (60 min)	×	3.5
30 minute		?

$$\frac{3.5}{2} = 1.75$$

$P[T > 1] =$ `1 - expon.cdf(x=1, scale=1/1.75)` $= 0.1739$

Log-Normal distribution

Consider an example of number of days of hospitalisation

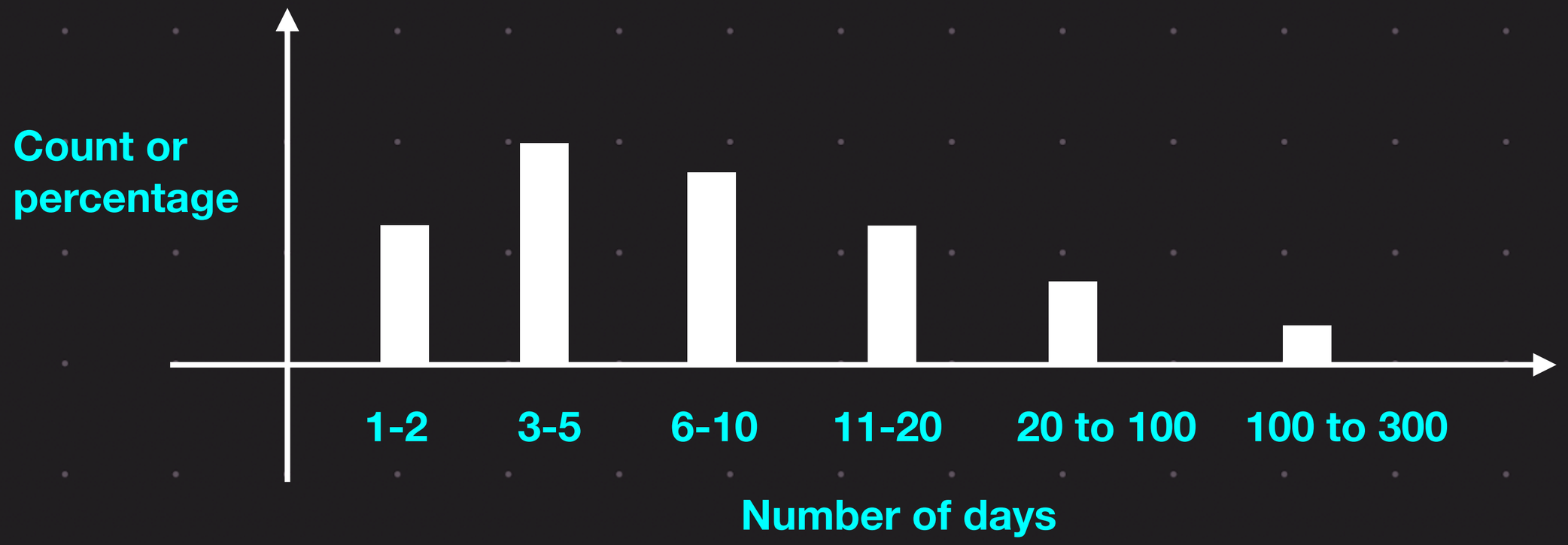
Most people may be hospitalised for 3 to 5 days

Quite often, just 1 or 2 days

But, there can be a few extreme cases in 20 to 30 days

Extremely rare cases of 300 days (coma etc.)

Log-Normal distribution

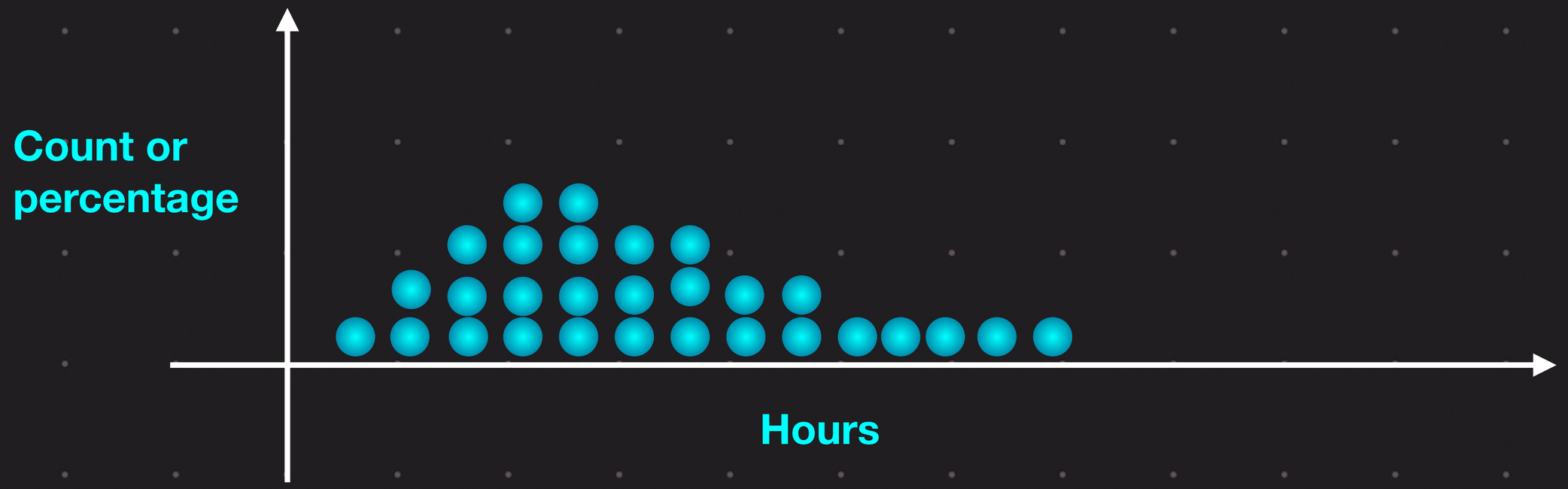


Number of days is discrete

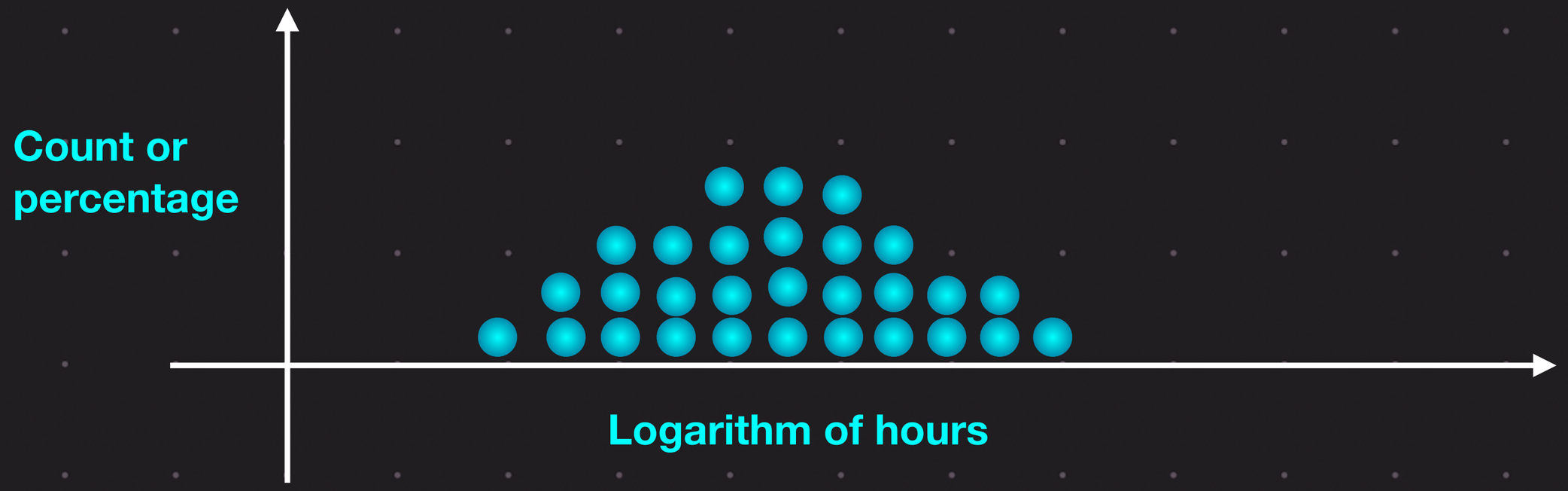
What if we had exact number of hours?

This would be continuous

Log-Normal distribution



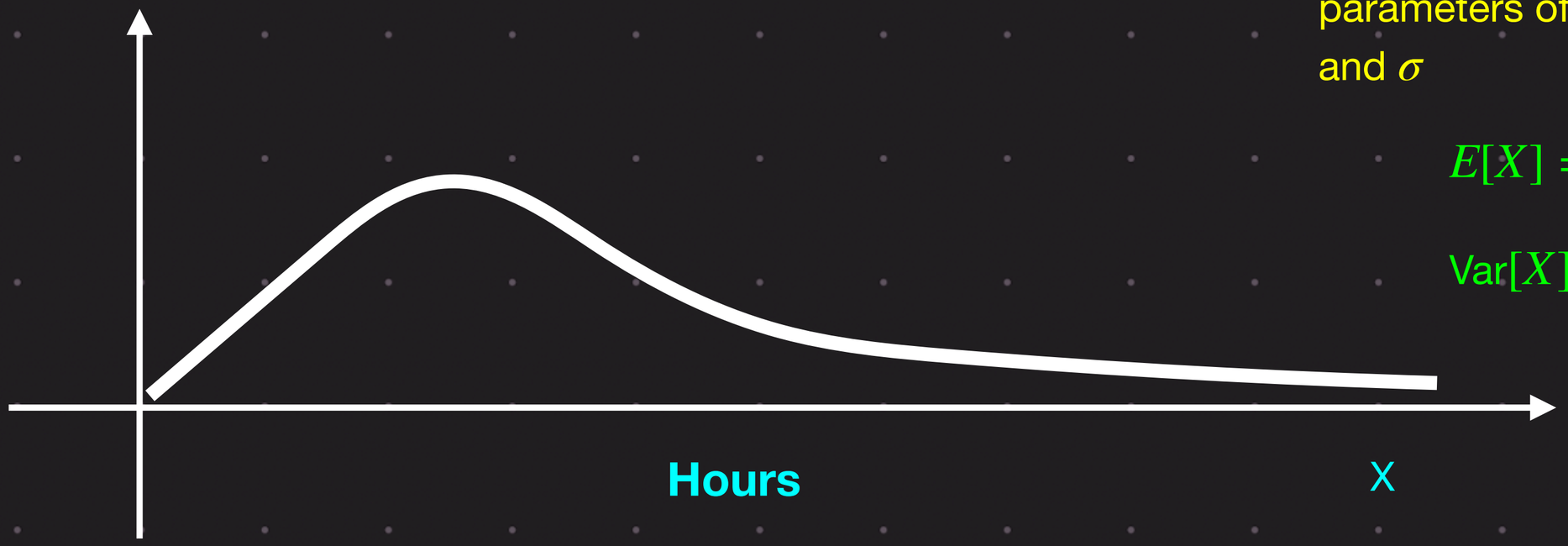
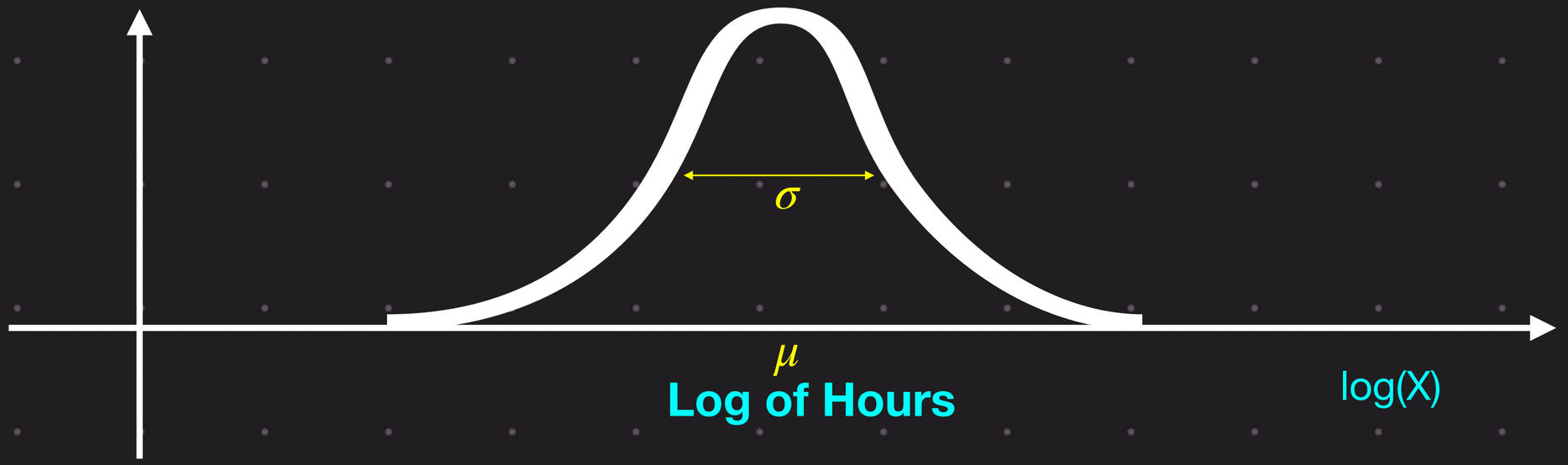
Log-Normal distribution



Log-Normal distribution

For the log normal, we use the parameters of the Gaussian: μ and σ

$$E[X] = e^{\mu+\sigma^2/2}$$

$$\text{Var}[X] = (e^{\sigma^2} - 1)\left(e^{2\mu+\sigma^2}\right)$$

Always confirm these equations from some source (eg: wiki)